

Task-based metacognition and insight in schizophrenia spectrum disorders: A systematic review and meta-analysis

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ABSTRACT

Impaired insight is prevalent among individuals with psychosis spectrum disorders and is associated with treatment non-adherence, symptom relapse, and functional decline. Metacognition has been proposed as a mechanism underlying insight impairment, but it is defined and measured in different ways, ranging from experimental tasks assessing confidence in perceptual decisions to broader self-reflective capacities. Whether behavioral metacognitive tasks are associated with insight remains unclear. We therefore conducted a systematic review to examine the association between behavioral metacognitive task performance and insight in schizophrenia spectrum disorders. We searched the PsycINFO, PubMed, Embase, and Scopus databases in accordance with PRISMA guidelines. Studies were included if they assessed adults with psychosis spectrum disorders using a behavioral metacognitive task and a validated insight measure. A random-effects meta-analysis with subgroup analyses by task type was conducted on studies reporting extractable correlations. Seventeen studies were included; only 6 (35.3%) reported significant associations. Across the 12 studies eligible for pooling, the overall metacognition–insight correlation was small but significant ($r = 0.18$, 95% CI [0.05, 0.31]; $I^2 = 60.2\%$). This effect varied substantially by task type: perceptual, probabilistic, metamemory, and social tasks showed no association with insight, whereas self-reflection paradigms ($r = 0.46$) and Meta-Wisconsin Card Sorting Test variants ($r = 0.33$) showed moderate associations. Task type accounted for approximately 89% of between-study heterogeneity. These results suggest that low-level, domain-specific metacognitive tasks do not capture processes relevant to insight, supporting hierarchical models of metacognition and the need for studies employing self-referential metacognitive tasks.

1. Introduction

Clinical insight impairment is among the most prevalent and clinically significant features of schizophrenia spectrum disorders, affecting an estimated 50%–80% of individuals with these conditions (Lincoln et al., 2007). Clinical insight refers to awareness of one's mental illness, including symptom recognition and acceptance of treatment need. Deficits in clinical insight are associated with reduced medication adherence, poorer engagement with psychiatric services, elevated relapse and rehospitalization rates, and diminished functional recovery (David, 2020). Distinct but related, Beck et al. (2004) introduced cognitive insight, capturing metacognitive processes involved in evaluating one's beliefs. Cognitive insight comprises two dimensions: self-

reflectiveness (openness to alternative interpretations of one's experiences), and self-certainty (overconfidence in one's conclusions). These dimensions are typically combined into a composite index (self-reflectiveness minus self-certainty), with higher scores reflecting a greater capacity to reflect on and revise one's beliefs relative to a tendency for overconfidence and resistance to corrective feedback (Beck et al., 2004). Notably, the studies included in the present review reported the composite index instead of the individual subscales. As such, the associations reported here are linked to overall cognitive insight and not specifically self-reflectiveness or self-certainty. Impaired cognitive insight has itself been associated with stronger delusional conviction, poorer response to treatment, and reduced psychosocial functioning (Van Camp et al., 2017).

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Clinical and cognitive insight are consistently positively correlated (Riggs et al., 2012), as both concern the capacity to reflect on and revise one's own beliefs, and they are therefore often measured together. Nonetheless, they appear to represent distinct constructs with different underlying mechanisms: the former related to specific attitudes and beliefs around mental illness and the latter seeking to capture a habitual cognitive style including but not specific to illness beliefs.

Researchers have proposed metacognition—the capacity to monitor, evaluate, and regulate one's own cognitive processes—as a mechanism underlying insight impairment (David et al., 2012; Flavell, 1979). Divergent operationalizations and assessment methods have emerged across cognitive neuroscience and clinical psychology (Heyes et al., 2020). Performance-based approaches quantify metacognition through behavioral paradigms yielding several metrics: metacognitive sensitivity, reflecting how well confidence discriminates between correct and incorrect responses; metacognitive bias, reflecting overall confidence independent of accuracy; and metacognitive efficiency, reflecting sensitivity adjusted for task performance (Fleming and Lau, 2014; Maniscalco and Lau, 2012).

In contrast, clinical approaches allow assessment of metacognition through self-report questionnaires and structured interviews, evaluating capacities such as self-reflectivity, understanding of others' mental states, decentration, and the application of metacognitive knowledge to manage psychological difficulties (Lysaker et al., 2018; Semerari et al., 2003). Whereas performance-based measures enable indexing of processes operating during discrete cognitive tasks, researchers use these instruments to assess generalized beliefs and capacities extending across situations (Seow et al., 2021). Consistent with this, several studies' authors have reported associations between metacognitive deficits and insight impairment in psychotic disorders (Bliksted et al., 2017; Chan, 2016). Beyond insight, metacognitive impairment carries broader clinical significance, having been linked to poorer functional outcomes in schizophrenia, independent of neurocognitive deficits (Davies and Greenwood, 2020).

Recently, Seow et al. (2025) proposed a hierarchical framework that organizes metacognitive processes across multiple levels, ranging from local, task-based monitoring to broader, self-referential evaluation of one's own beliefs and abilities. They highlight three levels of this framework that are particularly relevant to the present review, as the metacognitive paradigms used in the existing literature largely correspond to them. At the most local level, trial-level metacognition refers to moment to moment monitoring of performance on discrete cognitive operations and is typically assessed through the relationship between accuracy and confidence on perceptual, memory, or probabilistic reasoning tasks (Fleming and Lau, 2014; Maniscalco and Lau, 2012). This level is most closely aligned with confidence calibration and sensitivity (meta-d') traditionally examined in cognitive neuroscience. At a broader level, self-referential metacognition involves evaluating self relevant content, such as judging personal traits, reflecting on one's own characteristics, or integrating self related information across experiences, and is commonly assessed through paradigms requiring participants to evaluate, encode, or recall self related material (Northoff et al., 2006; Van der Meer et al., 2010). A third level concerns insight into one's own mental states and their disturbances. This encompasses cognitive insight, the recognition that one's beliefs may be biased or mistaken, as indexed by the self-reflectiveness and self-certainty subscales of the Beck Cognitive Insight Scale (Beck et al., 2004), as well as clinical insight, the capacity to recognize that one is ill, to relabel unusual experiences as pathological, and to appreciate the need for treatment (Amador et al., 1991; David, 1990). Both constructs require evaluating the validity of one's own beliefs and experiences, though clinical insight additionally demands that this evaluation be sustained across contexts and over time and judged against personal or social norms. Although these levels are conceptually distinct, they are likely interrelated. Trial-level metacognitive evaluations are known to affect more stable self-referential representations (Rouault et al., 2019;

Rouault and Fleming, 2020; Katyal et al., 2025), which in turn may shape awareness of one's beliefs and state of mind. Disruptions at any level may therefore influence metacognitive functioning in others.

However, synthesizing the relevant literature is complicated by substantial methodological heterogeneity. Researchers have employed diverse behavioral paradigms (e.g., perceptual discrimination, memory recognition, modified executive function tasks) and varied insight assessments (e.g., clinician-rated scales, self-report measures, cognitive insight instruments), limiting cross-study comparisons and the conclusions that can be drawn about this association (Hohendorf and Bauer, 2023; Hoven et al., 2019). To date, no researcher has conducted a systematic review to examine the specific relationship between performance-based metacognitive measures and insight in psychosis spectrum populations. This systematic review aims to examine the association between behaviorally measured metacognition and insight in individuals with schizophrenia spectrum disorders.

2. Methods

2.1. Protocol and registration

This systematic review was conducted in accordance with PRISMA 2020 guidelines (Page et al., 2021) and registered prospectively with PROSPERO (CRD420251090735).

2.2. Eligibility criteria

2.2.1. Population

Adults (≥ 18 years) with a primary diagnosis of schizophrenia spectrum or psychotic disorders per DSM (American Psychiatric Association, 2013) or ICD (World Health Organization, 2019) criteria, or individuals at clinical high risk for psychosis. Although clinical high-risk samples were eligible, no study meeting all inclusion criteria assessed an at-risk sample. Therefore, all studies in the final sample concerned established clinical populations.

2.2.2. Assessment

Studies were required to include: (1) objective task-based metacognitive assessments (e.g., confidence judgments, perceptual decision-making, metamemory tasks); (2) validated insight measures, including structured interviews (e.g., Scale to Assess Unawareness of Mental Disorder [SUMD]; Schedule for Assessment of Insight–Expanded [SAI-E]), observer-rated scales, self-rated scales (e.g., Beck Cognitive Insight Scale [BCIS]; Birchwood Insight Scale [BIS]), or single-item measures (e.g., Positive and Negative Syndrome Scale [PANSS] item G12); and (3) validated psychosis spectrum assessment. Studies were excluded if they only used self-report metacognition measures, those without validated insight measures, or those whose authors did not assess psychosis spectrum disorder.

2.2.3. Study design and restrictions

Original empirical studies in peer-reviewed journals using quantitative or mixed-methods designs were included. Reviews, theoretical papers, commentaries, editorials, letters, case reports, and non-peer-reviewed publications were excluded. Only published English-language studies were included; gray literature was excluded. No geographic restrictions were applied.

2.3. Information sources

A systematic search was conducted across four databases: PsycINFO (via APA PsycNet), PubMed, Embase (via Ovid), and Scopus between September 5–29, 2025, with no date restrictions, supplemented by Google Scholar. Reference lists of included articles were manually screened to identify additional eligible studies (citation tracking/“snowballing”).

2.4. Search strategy

Search terms were organized into three conceptual blocks combined with Boolean operators:

2.4.1. Block 1 (population)

Terms related to psychosis spectrum disorders, including schizophrenia*, schizoaffective, psychosis, “psychotic disorder*,” “first-episode psychosis,” FEP, “clinical high risk,” CHR*, UHR, “attenuated psychosis,” “at-risk mental state,” “prodromal psychosis,” and “delusional disorder.”

2.4.2. Block 2 (insight)

Terms related to clinical and cognitive insight, including “clinical insight,” “cognitive insight,” “insight into illness,” “illness awareness,” “self-awareness,” anosognosia, “self-reflectiveness,” “self-certainty,” as well as specific insight scale names (SUMD, BCIS, BIS, SAI, SAI-E, BABS, OVIS, MDIS, ITAQ, SAIQ, VAGUS, “PANSS insight,” “PANSS G12”).

2.4.3. Block 3 (metacognitive tasks)

Terms related to behavioral/experimental metacognitive assessments, including metacognition*, “metacognitive efficiency,” “metacognitive sensitivity,” “meta-d*,” “M-ratio,” “type 2 ROC,” “signal detection theory,” SDT, “confidence judgment*,” “confidence rating*,” “perceptual decision*,” “2AFC,” “random-dot motion,” “beads task,” “jumping to conclusions,” JTC, “probabilistic reasoning,” “draws to decision,” DTD, “metamemory task*,” “judgment of learning,” JOL,

“feeling of knowing,” FOK, “error monitoring,” “error awareness,” “performance monitoring,” and “post-decision wagering.”

Search terms were adapted for each database according to its specific syntax requirements (e.g.

field tags such as [ti,ab,kw] for PubMed)

2.5. Study selection

Records were exported and managed using Mendeley and Zotero, with duplicates identified and removed via DOI-based deduplication in Mendeley prior to import into Covidence. Covidence served as the primary platform for screening and data extraction. One reviewer (AH) screened titles and abstracts against eligibility criteria. One reviewer (AH) retrieved and assessed full-text articles, followed by verification by a second reviewer (AD).

2.6. Data extraction

Data were extracted from included studies using a standardized extraction form. The following information was extracted: (1) study characteristics (year of publication, sample size, and population); (2) participant characteristics (diagnosis, diagnostic criteria used, clinical subgroup); (3) metacognitive task details (task type, cognitive domain, outcome measures); (4) insight measures (scale type, specific instrument, and subscales); (5) symptom measures; and (6) main findings, including reported associations between metacognition and insight (e.g., correlation coefficients and statistical significance). A formal graded

Table 1

Characteristics of included studies, metacognitive tasks, insight measures, and key findings.

Study	Sample	N	Task domain	Specific task	Insight measure	Insight type	Key findings
Koren et al. (2004)	First-episode schizophrenia	30	Executive metacognition	Meta-WCST	SUMD	Clinical	$r = -0.41$ to -0.67^*
Kircher et al. (2007)	Schizophrenia	46	Metamemory	Metamemory recognition task	SUMD	Clinical	$r \approx 0.00$, ns
Bruno et al. (2012)	Schizophrenia	40	Executive metacognition	Emotional Meta-WCST	BCIS	Cognitive	$r = -0.34$ to -0.41^*
Van der Meer et al. (2012)	Schizophrenia	27	Self-reflection	Self-reflection fMRI task	SAI-E, BCIS	Clinical, Cognitive	$r = 0.48$ – 0.54^*
Bedford and David (2014)	Schizophrenia	26	Self-reflection	Levels-of-processing self-evaluation	SAI-E	Clinical	$r_s = 0.45$ – 0.56^*
De Vos et al. (2015)	Schizophrenia	37	Executive metacognition/self-reflection	ESRP, ISRP	SAI-E, BCIS	Clinical, Cognitive	$r = 0.30$ – 0.50^*
Moritz et al. (2015)	Schizophrenia	75	Probabilistic reasoning	Fish Task	BCIS	Cognitive	$F = 0.06$ – 0.37 , $p > 0.05$
Moritz et al. (2016)	Schizophrenia	80	Probabilistic reasoning	Fish Task	BCIS	Cognitive	$r = -0.02$ to 0.04 , ns
Marucci et al. (2019)	Schizophrenia	40	Executive metacognition	Meta-WCST adaptation	PANSS G12	Clinical	$r = 0.00$, ns
Alkan et al. (2020)	First-episode psychosis	49	Perceptual decision-making	Perceptual metacognition task	BCIS ^a	Cognitive	$r = -0.10$ to 0.07 , ns
Dietrichkeit et al. (2020)	Schizophrenia	59	Metamemory	VR-based metamemory task	BCIS	Cognitive	No direct association; paradoxical BCIS finding
Faivre et al. (2021)	Schizophrenia	41	Perceptual decision-making	Random-dot motion discrimination	BCIS, BIS ^b	Clinical, Cognitive	$p > 0.20$, ns
Palmer-Cooper et al. (2023)	Psychosis spectrum disorders	62	Perceptual decision-making	Meta-dots task	BCIS	Cognitive	$r \approx 0.00$, ns
Rouy et al. (2023)	Schizophrenia	38	Perceptual decision-making	Visual detection task	BCIS	Cognitive	$r = -0.08$ to 0.05 , ns
Roux et al. (2023)	Schizophrenia/schizoaffective	143	Social metacognition	V-COMICS adaptation	BIS, SUMD	Clinical	$\beta = 0.02$ – 0.03 , ns
Wright et al. (2024)	Schizophrenia spectrum disorders	41	Self-reflection	EMMP Task	BCIS ^a	Cognitive	$p = 0.01^*$
Espinosa et al. (2025)	First-episode psychosis	149	Probabilistic reasoning	Beads Task	BCIS, SUMD	Clinical, cognitive	Association not directly tested

Note. Meta-WCST = Metacognitive Wisconsin Card Sorting Test; BCIS = Beck Cognitive Insight Scale; SUMD = Scale to Assess Unawareness of Mental Disorder; SAI-E = Schedule for Assessment of Insight – Expanded Version; BIS = Birchwood Insight Scale; ISRP = Implicit Self-Related Processing Task; ESRP = Explicit Self-Related Processing Task; EMMP = Ecological Momentary Metacognitive Prediction Task; VR = Virtual Reality. All studies employed cross-sectional designs. N = sample size for clinical group.

ns = not significant.

* $p < 0.05$.

^a BCIS used as a metacognitive measure rather than as an insight measure.

^b BIS was measured but not analyzed in relation to metacognition.

risk-of-bias instrument was not applied because the included studies were uniformly cross-sectional and reported zero-order correlations rather than comparative or diagnostic effects. Instead, study quality was appraised using core indicators extracted for all studies (use of standardized diagnostic criteria, use of validated insight and behavioral metacognitive measures, and study precision reflected by sample size), which are summarized in Table 1.

The primary outcomes of interest were the reported associations between task-based metacognition and insight measures, and differences in metacognitive performance across clinical subgroups.

2.7. Data synthesis

A structured narrative synthesis was conducted, grouping studies by: (1) metacognitive task type (e.g., confidence judgments, perceptual decision-making tasks, metamemory tasks); (2) insight measure used (e.g., SUMD, SAI-E, BCIS, PANSS single-item); and (3) clinical population (schizophrenia, schizoaffective disorder). Correlation coefficients and key statistics were presented in tables organized by study characteristics. Patterns of associations between metacognition and insight were identified and compared across different assessment methods and populations.

To supplement the narrative synthesis, pooled effect size estimates were derived from studies reporting extractable Pearson correlations between behavioral metacognitive indices and insight measures. Effect sizes were pooled using a random effects model based on Fisher's z transformed correlations, with subgroup estimates calculated by task type. When studies reported multiple correlations across insight subscales or metacognitive indices, one representative correlation was selected per study to preserve independence, prioritizing the correlation most closely aligned with the primary metacognitive task category. For studies reporting a range of correlations, the midpoint of $|r|$ was used. This selection rule was specified a priori and applied uniformly to all studies, without reference to the magnitude, direction, or statistical significance of the resulting correlations. Consequently, effect-size selection could not have been guided by study outcomes. Clinical and cognitive insight were treated as a single insight construct because both fall under a broader umbrella of self-reflection, requiring individuals to evaluate the validity of their own beliefs and experiences, and because many studies assess the two together, with several of the included studies reporting both. Clinical and cognitive insight measures were also combined within the pooled and subgroup analyses because the primary subgrouping variable was metacognitive task type, for which only three studies were available per pooled subgroup. Further partitioning by insight type would have yielded subgroups of one to two studies, below the pre-specified threshold for pooling. Differences between clinical and cognitive insight are instead examined in the narrative synthesis.

For reverse coded insight measures (e.g., SUMD, denial scales), correlation signs were inverted so that positive values consistently reflected better metacognition associated with better insight. Fisher's z transformations were calculated as $z = \frac{1}{2} \ln[(1 + r)/(1 - r)]$, with variance estimated as $1/(N - 3)$, and pooled using the DerSimonian and Laird random effects estimator. Pooled estimates were then back transformed for interpretation. Heterogeneity was assessed using Cochran's Q , I^2 , and τ^2 . Subgroup analyses were conducted for task categories represented by at least three studies (perceptual decision making, Meta WCST variants, and self reflection paradigms), whereas smaller subgroups were summarized descriptively only. Between subgroup differences were examined using mixed effects meta regression, with R^2 used to estimate the proportion of heterogeneity attributable to task type. Potential small study effects were evaluated using funnel plots, Egger's regression, and trim and fill sensitivity analyses, alongside leave one out analyses for overall robustness. Of the 17 studies included in the narrative synthesis, five were excluded from pooled analyses due to insufficiently comparable effect size data but were retained narratively. Specifically, Faivre et al. (2021) and Moritz et al. (2015) did not

report reconstructable Pearson correlations, Dietrichkeit et al. (2020) did not report a direct metacognition–insight correlation, Espinosa et al. (2025) did not directly test this association, and Wright et al. (2024) reported cross lagged longitudinal coefficients rather than zero order correlations. Analyses were conducted in R using the metafor package (Viechtbauer, 2010).

3. Results

3.1. Study selection

A total of 26,703 references were imported for screening. Duplicate removal occurred in two stages, with 2115 duplicates identified manually and a further 1307 identified by Covidence, leaving 23,281 unique studies for title and abstract screening. Following title and abstract screening, 23,150 records were excluded and full-text articles were retrieved for 134 potentially eligible studies. After full-text review, 118 studies were excluded due to: no behavioral metacognitive task ($n = 74$); no validated insight measure ($n = 23$); not a peer-reviewed paper (e.g., conference poster or abstract; $n = 15$); not an empirical study (e.g., review, qualitative study; $n = 5$); and not published in English ($n = 1$). A total of 17 studies were included in the final synthesis, of which 15 were identified through database searches and 2 through citation tracking of included articles and relevant reviews. The selection process is illustrated in the PRISMA flow diagram (Fig. 1).

3.2. Study characteristics

The characteristics of the 17 studies are summarized in Table 1. All included adult participants (≥ 18 years) with schizophrenia spectrum disorders, including schizophrenia ($n = 10$), first-episode psychosis or schizophrenia ($n = 3$), schizophrenia/schizoaffective disorder ($n = 1$), and psychosis spectrum disorders ($n = 2$). None of the included studies assessed a clinical high-risk or at-risk mental state sample; all samples were comprised of established psychosis spectrum populations, predominantly chronic schizophrenia or mixed psychosis spectrum groups, with relatively few studies in any single diagnostic category. Overall study quality was moderate to high: all the included studies used standardized diagnostic criteria and validated insight and metacognitive measures, the latter two being inclusion requirements. The studies varied primarily in their precision; under random-effects weighting, smaller studies contributed proportionally less to the pooled estimates, and leave-one-out analysis confirmed that no single study disproportionately influenced the results. Study quality did not appear to correlate systematically with effect size.

3.3. Metacognitive approaches

3.3.1. Experimental/task-based vs. self-reflective approaches

Two distinct approaches to measuring metacognition were identified across the included studies. The authors of the majority ($n = 14$) employed experimental metacognition paradigms assessing local metacognitive processes through the correspondence between objective performance and subjective confidence judgments. A smaller subset ($n = 3$) employed self-reflective paradigms assessing self-referential processing (Bedford and David, 2014; De Vos et al., 2015; Van der Meer et al., 2012), while two author groups used metamemory or ecological momentary assessment tasks (Kircher et al., 2007; Wright et al., 2024).

3.3.2. Specific metacognitive tasks

3.3.2.1. Perceptual decision-making tasks with confidence judgments. Four studies' authors employed perceptual decision-making tasks paired with confidence ratings. Rouy et al. (2023) used a visual detection task in which participants judged whether a face was present in noisy images,

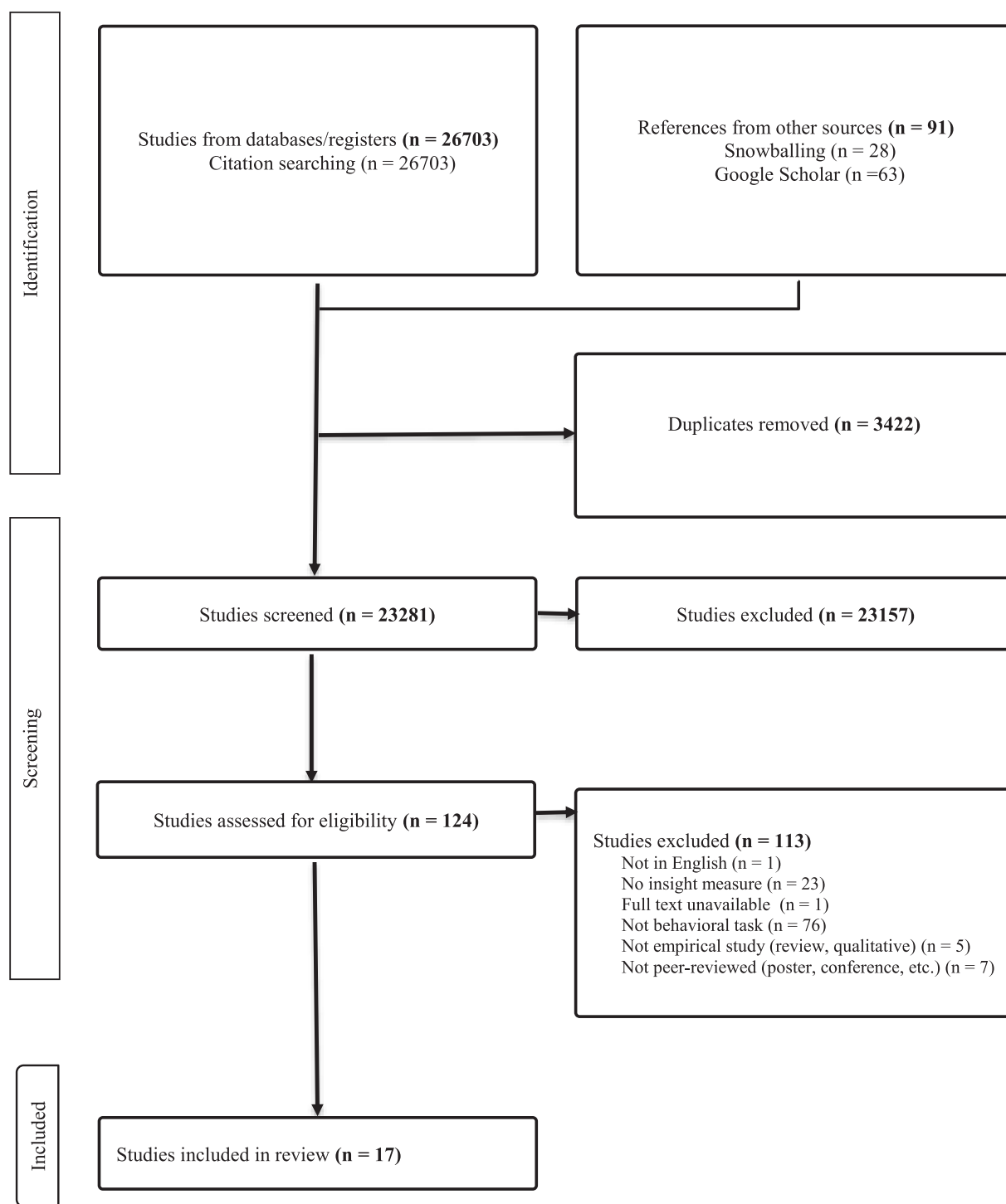


Fig. 1. PRISMA 2020 flow diagram for new systematic review.

followed by confidence ratings on a continuous scale. Faivre et al. (2021) employed a random-dot kinematogram in which participants judged motion direction and rated confidence. Palmer-Cooper et al. (2023) used the meta-dots task, requiring participants to judge which side of a display contained more dots and rate their confidence. Alkan et al. (2020) employed a perceptual metacognitive accuracy task adapted from Fleming et al. (2010, 2012). These paradigms measure metacognitive sensitivity through signal detection theory-based metrics (e.g., meta- d' /d').

3.3.2.2. *Probabilistic reasoning tasks (Beads Task/Fish Task)*. Three studies' authors employed the Beads Task to assess metacognitive aspects of probabilistic reasoning. Moritz et al. (2015, 2016) used the Fish Task, a version of the Beads in which participants observe fish drawn sequentially from one of two lakes with different probability ratios (80/20) and decide which lake the fish are coming from. Espinosa et al. (2025) employed the Beads Task across three conditions: 85/15, 60/40, and an affective 60:40 in which participants drew positive and negative self-referent adjectives instead of beads. Across all studies, the primary outcome was draws to decision, with jumping to conclusions (JTC) defined as deciding after one or two draws, indexing the

calibration of confidence in probabilistic reasoning. [Espinosa et al. \(2025\)](#) is included because they provided relevant data on both constructs across a large first-episode psychosis sample ($N = 149$), though they did not test the direct association between metacognition and insight.

3.3.2.3. Metacognitive Wisconsin Card Sorting Test (Meta-WCST) variants. Four studies' authors employed variants of the Meta-WCST ([Koren et al., 2004](#)), which extends the traditional executive-function test by adding confidence ratings and voluntary response decisions. After each sorting decision, participants rated their confidence from 0 ("just guessing") to 100 ("completely certain"), then chose whether to "volunteer" that response toward their total score, earning or losing a small monetary reward depending on accuracy. This yields indices of metacognitive monitoring (accuracy of confidence judgments), control sensitivity (the relation between confidence and decision to volunteer), and free-choice improvement. [Bruno et al. \(2012\)](#) developed an Emotional Meta-WCST in which participants sorted 64 cards displaying emotionally neutral, positive, or negative words, rating confidence and deciding whether to volunteer each response providing a measure of metacognitive awareness under emotionally salient conditions. [Marucci et al. \(2019\)](#) employed a similar adaptation. [De Vos et al. \(2015\)](#) used an explicit self-related processing task (ESRP) adapted from the Meta-WCST, in which participants sorted cards, rated confidence, and decided whether to include responses before receiving feedback, with the option to revise their decision afterwards; indices captured metacognitive monitoring and control.

3.3.2.4. Social metacognition tasks. [Roux et al. \(2023\)](#) employed an adapted social metacognition task derived from the Versailles-Cognition in Mental Illness and Cognitive Sciences (V-COMICS) test, in which participants interpreted social interactions depicted in comic strips and rated their confidence in mentalizing judgments. This task measures metacognitive monitoring in the context of the theory of mind and social-cognitive processing.

3.3.2.5. Self-reflection and metamemory tasks. Six studies' authors employed self-reflection or metamemory paradigms. [Van der Meer et al. \(2012\)](#) used a self-reflection fMRI task adapted from [Modinos et al. \(2009\)](#), in which participants rated their agreement with self-referential statements while the researchers measured neural activation. The self- and other-related conditions required participants to reflect on personal or interpersonal traits (self-reflective metacognition), while the factual statements served as control. [De Vos et al. \(2015\)](#) employed two paradigms: the Implicit Self-Related Processing Task (ISRP) and the ESRP. In the ISRP, participants read scrambled sentences containing either positive or negative self-descriptive words and then completed a brief working-memory task. Accuracy and speed on these primes reflected automatic, implicit self-processing. [Wright et al. \(2024\)](#) employed the Ecological Momentary Metacognitive Prediction Task (EMMP), implemented via smartphone. They prompted participants several times per day across a 2-week period to identify a real-world activity they planned to complete (e.g., making a meal, studying, contacting someone). Predicted versus actual responses indexed momentary metacognitive accuracy in daily life.

[Dietrichkeit et al. \(2020\)](#) employed a virtual reality (VR)-based metamemory paradigm in which participants explored VR environments (e.g., a street scene, a metro station, a beach, a campground) and subsequently completed a recognition task for faces and objects. The researchers computed the Knowledge Corruption Index (KCI) as the ratio of incorrect high-confident responses to all high-confident responses. They also employed a randomized-controlled design with corrective performance feedback to examine effects on overconfidence and cognitive insight.

[Bedford and David \(2014\)](#) used a levels-of-processing paradigm to

examine self-reflection depth in schizophrenia. Participants made judgments about the self-descriptiveness of mental illness traits, negative traits, and positive traits, followed by surprise recall and recognition tests. Unlike confidence-based paradigms, this task operationalizes metacognitive processing through episodic memory performance, based on the theory that deeper self-reflective processing during encoding results in better retrieval. In contrast, [Kircher et al. \(2007\)](#) used a metamemory recognition task to assess confidence in memory judgments for neutral rather than self-referential stimuli.

3.4. Insight measures

All metacognitive task and insight measure combinations and findings are presented in [Table 1](#).

3.4.1. Cognitive insight measures

The BCIS ([Beck et al., 2004](#)) was the most frequently used cognitive insight measure ($n = 12$ studies). It is a 15-item self-report scale comprising two subscales: self-reflectiveness (openness to alternative explanations) and self-certainty (overconfidence in beliefs). Researchers typically compute a composite index (self-reflectiveness minus self-certainty) as an overall indicator of cognitive insight. Two studies' authors used the BCIS as a measure of metacognitive capacity rather than insight ([Alkan et al., 2020](#); [Wright et al., 2024](#)).

3.4.2. Clinical insight measures

Authors of four studies employed the SUMD ([Amador et al., 1993](#)), assessing awareness of disorder, medication effects, and social consequences. [Birchwood et al. \(1994\)](#) used the BIS, and [Kemp and David \(1997\)](#) used the SAI-E. [Kay et al. \(1987\)](#) used PANSS item G12 as a single-item insight measure.

3.5. Synthesis of findings

3.5.1. Overall pattern of results

Of the 17 studies, authors of 6 (35.3%) reported significant associations between task-based metacognition and insight ($r = 0.30$ – 0.67), whereas authors of 10 (58.8%) found no significant relationship ($r = -0.10$ – 0.07). Authors of one study ([Espinosa et al., 2025](#)) assessed both task-based metacognition and insight but did not directly test their association. The pattern of findings varied based on metacognitive task type and insight measure employed.

3.5.2. Metacognitive task type

3.5.2.1. Perceptual decision-making tasks. None of the four studies' authors who used perceptual decision-making tasks with confidence judgments (i.e., visual detection, random-dot motion discrimination, meta-dots) found significant associations with insight measures. These tasks, which allow assessment of metacognitive sensitivity in low-level perceptual processing, showed near-zero correlation with both cognitive insight (BCIS) and clinical insight (BIS) ($r = -0.10$ – 0.07 ; all $p > 0.20$).

3.5.2.2. Probabilistic reasoning tasks. Neither of the two studies' authors who used the Fish Task/Beads Task found significant relationships between JTC indices or confidence calibration and cognitive insight (BCIS; $r = -0.02$ – 0.04). [Espinosa et al. \(2025\)](#) assessed JTC bias (Beads Task) and insight (BCIS, SUMD) in 149 individuals with first-episode psychosis. Although they did not directly test the association, cluster analysis revealed that neuropsychological profiles differing in clinical insight (SUMD) did not differ in JTC performance or cognitive insight (BCIS), suggesting potential independence between task-based metacognition and insight.

3.5.2.3. Metamemory tasks. Two studies' authors employed metamemory paradigms. Kircher et al. (2007) used a metamemory recognition task assessing confidence in memory for neutral stimuli and found no significant correlation between metamemory performance and clinical insight (SUMD) ($r \approx 0.00$, ns). Dietrichkeit et al. (2020) employed a VR-based metamemory paradigm in which participants completed a recognition task for faces and objects, rating their confidence. Patients with psychosis displayed overconfidence in false memories for emotions and gave significantly more high-confident responses than healthy controls. Paradoxically, patients scored significantly higher than controls on the BCIS composite index ($t(57) = 2.09$, $p = 0.041$), which indicates greater self-reported cognitive insight despite objective overconfidence. This composite difference reflected significantly higher scores among patients on both subscales—self-reflectiveness ($t(57) = 4.33$, $p < 0.001$) and self-certainty ($t(54.48) = 2.22$, $p = 0.030$)—with the larger self-reflectiveness difference driving the composite. That patients simultaneously reported greater self-reflectiveness and greater self-certainty while displaying objective overconfidence underscores the dissociation between self-reported cognitive insight and behaviorally measured metacognition. Corrective feedback did not improve cognitive insight. Neither group found a significant association between task-based performance and insight.

3.5.2.4. Meta-WCST variants. Results were mixed, with three of four studies' authors reporting significant associations. Koren et al. (2004) reported that metacognitive indices (free-choice improvement, control sensitivity) were strongly correlated with clinical insight (SUMD; $r = -0.41$ to -0.67), explaining substantially more variance than standard cognitive measures. Bruno et al. (2012) found that metacognitive monitoring on an emotional variant was associated with cognitive insight (BCIS self-certainty: $r = -0.34$ to -0.41). However, Marucci et al. (2019) found no relationship between Meta-WCST performance and insight (PANSS iG 12).

3.5.2.5. Self-reflection tasks. All three studies' authors who employed self-reflection paradigms found significant associations with insight. Van der Meer et al. (2012) reported that neural activation during self-reflection correlated with both clinical (SAI-E) and cognitive insight (BCIS). De Vos et al. (2015) found significant relationships between implicit and explicit self-related processing and both insight measures. Bedford and David (2014) found that lower recall for self-evaluated mental illness traits significantly correlated with higher denial of illness (SAI-E). Across the three studies, associations ranged from moderate to large ($r/r_s = 0.30$ – 0.56). Notably, this team measured self-reflection through memory performance rather than confidence ratings, suggesting that self-referential processing and insight associations extend beyond confidence-based paradigm.

3.5.3. Insight measure

3.5.3.1. Cognitive insight (BCIS). Of the 10 studies whose authors used the BCIS, authors of three reported significant associations (30%; $r = 0.30$ – 0.54). Significant findings emerged with Meta-WCST variants and self-reflection tasks. Researchers using perceptual decision-making tasks found near-zero correlation with BCIS ($r = -0.08$ – 0.05). Dietrichkeit et al. (2020) found no association between VR-based metamemory and BCIS despite objective overconfidence. Espinosa et al. (2025) found no significant differences in BCIS scores across neuropsychological profiles that differed in clinical insight, though they did not test the direct metacognition–insight association.

3.5.3.2. Clinical insight (SUMD, SAI-E, BIS, PANSS G12). Of five studies which examined clinical insight, two found significant associations (40%; $r/r_s = 0.30$ – 0.67). Significant findings emerged with Meta-WCST (SUMD; $r = -0.41$ to -0.67) and self-reflection tasks (SAI-E; $r/r_s =$

0.30 – 0.56). Researchers using social metacognition and the PANSS item found no significant relationships.

3.6. Pooled effect-size synthesis

Pooled estimates were derived from 12 of the 17 included studies for which a representative metacognition–insight Pearson correlation could be extracted (N range: 26–143). The remaining five studies were retained in the narrative synthesis only: Faivre et al. (2021) and Moritz et al. (2015) reported only inferential statistics from which a Pearson r could not be reconstructed; Dietrichkeit et al. (2020) reported no direct correlation between the metamemory task and insight; Espinosa et al. (2025) did not directly test the metacognition–insight association; and Wright et al. (2024) reported only longitudinal cross-lagged regression coefficients rather than a zero-order correlation. The overall pooled correlation was small but statistically significant, $r = 0.18$ (95% CI $[0.05, 0.31]$, $p = 0.008$), with substantial between-study heterogeneity ($Q(11) = 27.64$, $p = 0.004$, $I^2 = 60.2\%$, $\tau^2 = 0.032$). Per-study effect-size estimates used in the pooled synthesis are listed in Table 2; subgroup-level summaries and heterogeneity statistics are presented in Table 3, and the corresponding forest plot is shown in Fig. 2.

Subgroup estimates, computed for task types with $k \geq 3$ studies, indicated that the overall pooled effect was driven by tasks engaging higher-order, integrative metacognitive processes. Perceptual decision-making tasks ($k = 3$) yielded a pooled correlation indistinguishable from zero ($r = -0.01$, 95% CI $[-0.17, 0.16]$, $p = 0.95$; $I^2 = 0.0\%$), whereas self-reflection paradigms ($k = 3$) yielded a moderate and statistically significant pooled correlation ($r = 0.46$, 95% CI $[0.27, 0.61]$, $p < 0.001$; $I^2 = 0.0\%$). Meta-WCST variants ($k = 3$) showed an intermediate, marginally significant effect with high heterogeneity ($r = 0.33$, 95% CI $[-0.01, 0.61]$, $p = 0.058$; $I^2 = 70.1\%$). A mixed-effects meta-regression with task type as moderator was highly significant, $QM(2) = 12.04$, $p = 0.002$, and task type accounted for approximately 89% of the between-study heterogeneity ($R^2 \approx 89\%$), indicating that the majority of the variability in observed metacognition–insight correlations was attributable to differences in the metacognitive paradigm employed. Subgroups with fewer than three studies (probabilistic reasoning, metamemory, social metacognition) yielded near-zero correlations descriptively (r values: 0.01, 0.00, and 0.03 respectively) but were not pooled.

Funnel-plot asymmetry was significant on Egger's regression test ($t(10) = 2.90$, $p = 0.016$). Inspection of the funnel plot (Fig. 3) indicates that this asymmetry is largely attributable to the systematic relationship between task type and effect size: studies employing self-reflection and Meta-WCST paradigms cluster on the high-effect side, while studies

Table 2
Per-study effect sizes used in the pooled synthesis ($k = 12$).

Study	r	Task type
Koren et al. (2004)	+0.54	Meta-WCST
Kircher et al. (2007)	0.00	Metamemory
Bruno et al. (2012)	+0.43	Meta-WCST
Van der Meer et al. (2012)	+0.51	Self-reflection
Bedford and David (2014)	+0.49	Self-reflection
De Vos et al. (2015)	+0.39	Self-reflection
Moritz et al. (2016)	+0.01	Probabilistic reasoning
Marucci et al. (2019)	0.00	Meta-WCST
Alkan et al. (2020)	−0.01	Perceptual decision-making
Palmer-Cooper et al. (2023)	0.00	Perceptual decision-making
Rouy et al. (2023)	0.00	Perceptual decision-making
Roux et al. (2023)	+0.03	Social metacognition

Note. r represents the metacognition–insight correlation extracted for each study included in the pooled analysis. When multiple correlations were reported, the estimate corresponding to the study's primary metacognitive task was selected. For reverse-coded insight measures, correlation signs were inverted so that positive values indicated better metacognitive performance associated with better insight. $k = 12$.

Table 3
Pooled effect-size estimates by metacognitive task type.

Subgroup	k	Pooled r	95% CI	p	I ²
Overall pooled	12	+0.18	[0.05, 0.31]	0.008	60.2%
Perceptual decision-making	3	−0.01	[−0.17, 0.16]	0.953	0.0%
Meta-WCST variants	3	+0.33	[−0.01, 0.61]	0.058	70.1%
Self-reflection paradigms	3	+0.46	[0.27, 0.61]	<0.001	0.0%
Probabilistic reasoning	1	+0.01	–	–	–
Metamemory	1	0.00	–	–	–
Social metacognition	1	+0.03	–	–	–

Note. Pooled correlations were estimated using a DerSimonian–Laird random-effects model on Fisher’s z-transformed correlations and back-transformed for presentation. Pooled estimates and heterogeneity statistics are reported only for subgroups with $k \geq 3$ studies; subgroups with $k = 1$ are reported descriptively. Test of task type as a moderator: $QM(2) = 12.04, p = 0.002$; task type accounted for approximately 89% of between-study heterogeneity ($R^2 \approx 89\%$). k = number of studies; 95% CI = 95% confidence interval; I^2 = proportion of total variability attributable to between-study heterogeneity.

employing perceptual, probabilistic, social, and metamemory tasks cluster around zero. This task-driven heterogeneity is itself the substantive finding of the review rather than a clear publication-bias signal, although both interpretations cannot be definitively separated with the

present sample. A trim-and-fill sensitivity analysis imputed 2 potentially missing studies on the left side; the adjusted pooled estimate was $r = 0.12$ (95% CI $[-0.02, 0.26]$, $p = 0.09$), suggesting that the observed pooled estimate could be inflated under a strict publication-bias interpretation, although the asymmetry is primarily explained by paradigm-related heterogeneity. A leave-one-out sensitivity check confirmed that the overall pooled estimate was robust to the removal of any single study (range of pooled r : 0.14 to 0.20), with no individual study qualitatively reversing the conclusion.

4. Discussion

This systematic review examined the relationship between task-based metacognition and insight in schizophrenia spectrum disorders. Across 17 studies, findings revealed an inconsistent pattern, with only 35.3% of studies showing significant associations ($r/r_s = 0.30$ – 0.67). The results suggest that the relationship between metacognition and insight critically depends on how metacognition is operationalized. Studies whose authors employed perceptual decision-making tasks consistently showed near-zero correlation with insight. Similarly, probabilistic reasoning tasks showed no relationship with cognitive insight. [Espinosa et al. \(2025\)](#) further found that, in first-episode psychosis, groups differing in clinical insight did not differ in JTC performance, providing no evidence of an association between JTC bias and

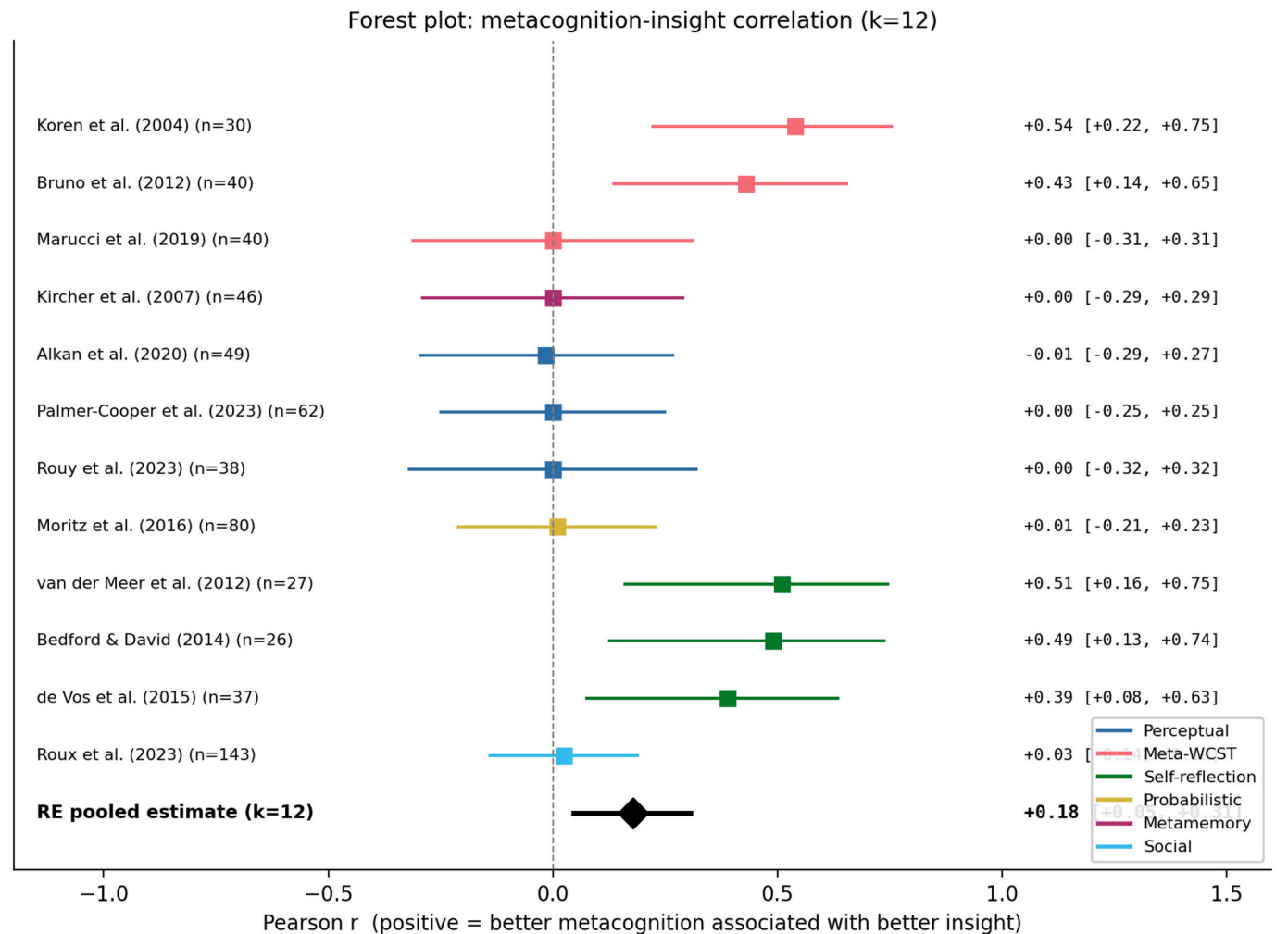


Fig. 2. Forest plot of metacognition–insight Pearson correlations across the 12 studies included in the pooled synthesis ($k = 12$), grouped by metacognitive task type. Effect sizes are presented as point estimates with 95% confidence intervals. Subgroup pooled estimates (perceptual decision making, Meta-WCST variants, and self reflection paradigms), as well as the overall pooled estimate, are displayed as summary diamonds. Positive values indicate that better metacognition was associated with better insight.

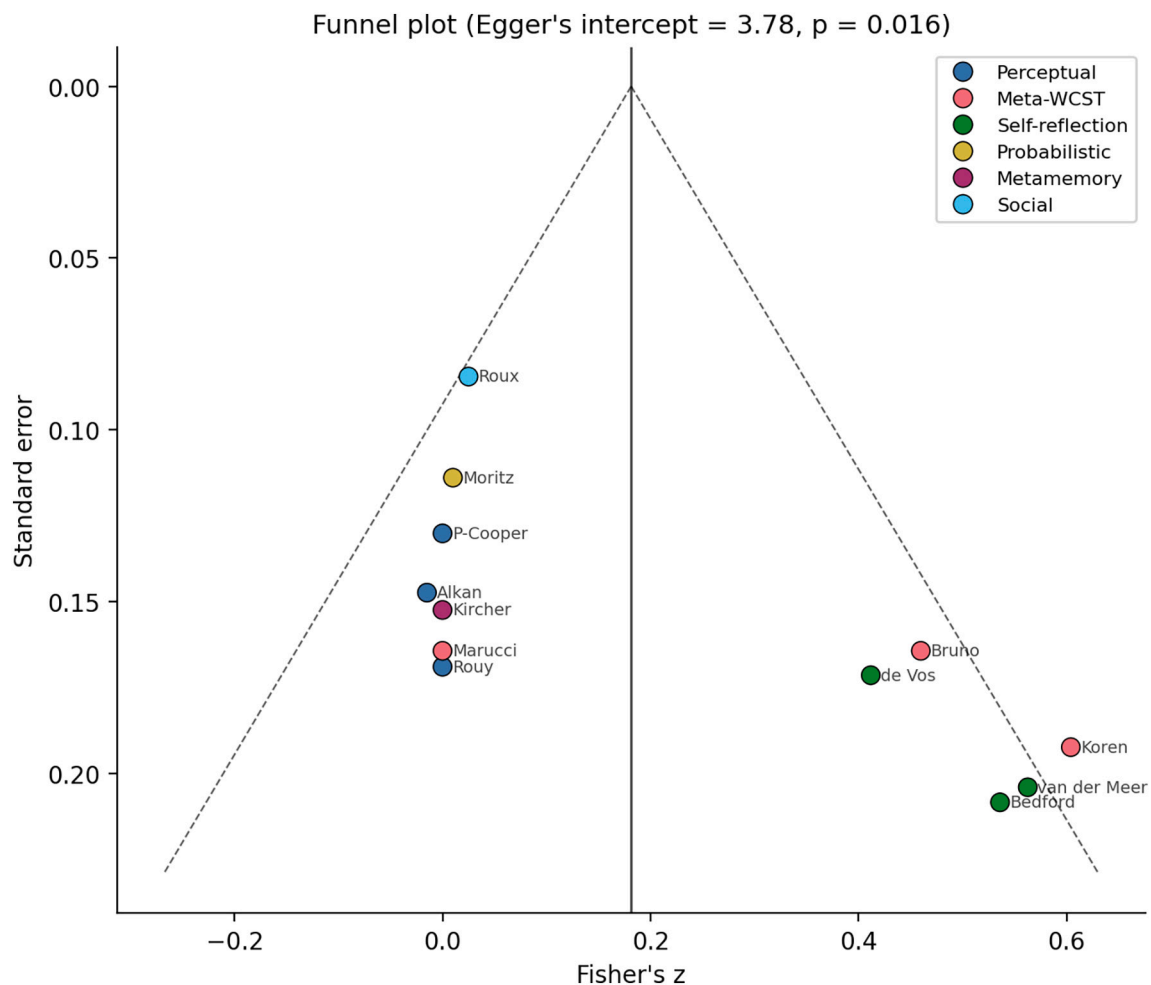


Fig. 3. Funnel plot of the 12 effect sizes included in the pooled synthesis, plotting Fisher's z transformed correlations against standard error. The vertical line represents the overall pooled estimate ($r = 0.18$). Egger's regression test indicated significant asymmetry ($t(10) = 2.90$, $p = 0.016$). Visual inspection suggests that this asymmetry may partly reflect systematic differences in effect sizes across metacognitive task types. A trim and fill sensitivity analysis imputed two potentially missing studies on the left side, yielding an adjusted pooled estimate of $r = 0.12$ (95% CI $[-0.02, 0.26]$).

insight.

In contrast, tasks engaging higher-order cognitive processes showed more consistent associations with insight. Meta-WCST variants yielded mixed results, with two of four studies highlighting significant correlations. Notably, authors of all three self-reflection studies reported significant associations with insight measures, including Bedford and David (2014), who used memory-based rather than confidence-based indices, demonstrating that this relationship extends beyond explicit metacognitive judgments. This pattern suggests that metacognition is not a unitary construct (Seow et al., 2021, 2025). Global metacognitive processes involving self-referential thinking appear to share more variance with insight, consistent with theories that insight requires the ability to reflect on one's own mental states and experiences (Van der Meer et al., 2012).

These findings point to a dissociation: individuals with psychosis spectrum disorders may demonstrate preserved metacognitive accuracy on perceptual tasks yet exhibit impaired awareness of their psychiatric condition. This suggests that insight requires more than confidence calibration. Instead, it may depend on the integrating self-relevant information across time and contexts, forming a coherent representation of one's experiences and patterns of functioning. Tasks assessing local metacognitive processes do not appear to capture this integrative, self-referential capacity.

The pooled effect-size synthesis reinforces and sharpens the qualitative pattern. Across the 12 studies eligible for pooling, the overall

pooled correlation between task-based metacognition and insight was small but significant ($k = 12$, $r = 0.18$, 95% CI $[0.05, 0.31]$). However, this aggregate estimate masked substantial task-related heterogeneity ($I^2 = 60.2\%$). When the metacognitive paradigm involved only local, trial-level monitoring (i.e., perceptual decision-making, probabilistic reasoning, metamemory, and social metacognition), the pooled or single-study association with insight was indistinguishable from 0 (e.g., perceptual decision-making, $k = 3$, $r = -0.01$). When the paradigm involved self-referential or executive-metacognitive processing, pooled correlations were moderate (self-reflection, $k = 3$, $r = 0.46$; Meta-WCST variants, $k = 3$, $r = 0.33$), with the self-reflection subgroup yielding a particularly homogeneous estimate. Meta-regression indicated that task type accounted for the majority of between-study heterogeneity (approximately 89%), suggesting that once paradigm type is taken into account, the studies are nearly homogeneous in their estimates. This pattern provides quantitative support for the hierarchical interpretation that insight in psychosis is more closely tied to higher-order, integrative self-monitoring than to trial-level confidence calibration (Seow et al., 2025).

This pattern suggests that insight impairment in psychosis may reflect dysfunction at global rather than local levels of metacognitive processing, consistent with the hierarchical organization of metacognition (Seow et al., 2025). Within this framework, clinical insight appears most closely tied to higher-order processes of self-referential evaluation rather than to the local, domain-specific metacognitive processes such as

trial-level confidence in perceptual judgments.

This review primarily contrasts the extremes of the metacognitive hierarchy, local domain-specific metacognitive monitoring (e.g., perceptual, metamemory, and probabilistic reasoning tasks) versus global self-referential processing. Intermediate levels, such as domain-level self-beliefs, social-cognitive monitoring, and awareness of cognitive biases (Seow et al., 2025), remain underexplored in relation to insight. Examining these levels could clarify how local metacognitive computations scale up to global self-evaluations. For example, social metacognition (e.g., Roux et al., 2023) and executive metacognitive monitoring (e.g., Meta-WCST studies) may represent such intermediate processes, partly accounting for their mixed findings.

An important open question is how trial-level metacognition relates dynamically to self-referential metacognition. The fact that perceptual and self-reflection tasks show distinct associations with insight does not mean that these levels are causally independent. Recent work by Katyal et al. (2025) provides one possible mechanism through which local and global metacognition may become coupled. In healthy individuals, global self-performance estimates were systematically updated by local confidence during trial-by-trial tasks, and asymmetries in external feedback further shifted affective self-beliefs. Critically, this mechanism appeared distorted in individuals with elevated subclinical anxious depression symptoms, who showed reduced sensitivity of global confidence to instances of higher local confidence despite intact task performance. If a similar disruption in the local to global updating process occurs in psychosis, it could help explain how trial-level metacognitive function may remain preserved, as seen in the perceptual subgroup of the present review, while higher-order self-evaluation, and consequently insight, remains impaired. The present review cannot determine this hypothesis, as the included studies assessed metacognition at one hierarchical level at a time rather than examining integration across levels. Future studies that assess local and global metacognition within the same participants, ideally alongside experimental manipulations of feedback, are needed to test whether local to global mapping is selectively disrupted in psychosis and whether such disruption contributes to insight impairment.

Whether metacognition relates differentially to clinical versus cognitive insight is a key question. Clinical insight significantly correlated with metacognition in 40% of studies, whereas cognitive insight showed significant associations in 30% of studies. This may reflect the nature of the constructs being assessed: clinical insight involves recognizing disorder and need for treatment, requiring global self-monitoring capacities (Lysaker et al., 2011; Seow et al., 2025), whereas cognitive insight may capture aspects of metacognition distinct from the trial-by-trial confidence calibration assessed by perceptual tasks. The significant findings for Meta-WCST and self-reflection tasks support this interpretation because these paradigms assess metacognitive monitoring requiring integration of self-relevant information. Cognitive insight, by contrast, reflects a broader capacity to evaluate one's beliefs that is not necessarily illness-specific and is less directly captured by behavioral tasks, particularly the trial-by-trial confidence calibration assessed by perceptual paradigms. Consistent with this, the near-zero correlations between perceptual metacognitive efficiency and BCIS scores suggest that different levels of self-monitoring are being assessed (Lehmann et al., 2022; Palmer-Cooper et al., 2023). Several features of the cognitive insight construct may explain this divergence. Trial-by-trial metacognitive indices (e.g., meta-d') quantify the online, stimulus-bound coupling between confidence and accuracy on a single class of discrete decisions. In contrast, the BCIS indexes trait-level, retrospective beliefs about one's reasoning that are intended to generalize across situations rather than track performance on any particular task. There is thus little reason to expect self-reflectiveness—a dispositional openness to feedback and alternative interpretations—to correlate with trial-level metacognitive sensitivity because it is conceptually closer to global self-evaluation than to moment-to-moment monitoring. Self-certainty, meanwhile, reflects overconfidence in high-level beliefs rather than

miscalibration on a specific stimulus, and elevated belief-level certainty can coexist with intact local metacognitive sensitivity. Because the composite index subtracts these two partially dissociable dimensions, any dimension-specific relationship with behavioral metacognition may be further attenuated or cancelled at the composite level. Consistent with dimensional asymmetry, a prior meta-analysis found that self-certainty was more consistently associated with objective cognitive performance than self-reflectiveness (Nair et al., 2014). Therefore, aggregating the two into a single composite may obscure a dimension-specific link to task-based metacognition. Finally, the BCIS relies on introspective self-reporting rather than behaviorally derived indices, a measurement-level divergence consistent with evidence that individuals with psychosis may not accurately estimate their own metacognitive functioning (Moritz et al., 2016). Dietrichkeit et al. (2020) further found paradoxically higher BCIS scores in patients despite overconfidence in false memories, suggesting that self-report cognitive insight measures may not accurately capture metacognitive functioning in psychosis. This is consistent with evidence that patients may overestimate their cognitive insight (Moritz et al., 2016). If BCIS scores do not correspond with objective performance, reliance on the BCIS alone may obscure metacognition–insight associations detectable using behaviorally derived indices.

These findings have several implications for clinical practice and future research. First, there is a need for developing and validating behavioral tasks that allow assessment of self-referential metacognitive processes, given most existing paradigms measure domain-specific metacognition, which shows no consistent associations with insight, whereas self-referential and executive metacognitive tasks do. Future behavioral tasks should engage higher-order self-evaluative processes rather than domain-specific confidence monitoring. Second, the dissociation between different types of metacognitive tasks highlights the need for careful task selection in insight research. Perceptual metacognition paradigms may not capture the self-reflective processes most relevant to insight. Third, if insight impairment reflects deficits in global self-reflection rather than domain-specific metacognitive monitoring, interventions should prioritize self-reflective capacities, focusing on helping individuals develop coherent narratives about their experiences, recognize patterns in their functioning, and integrate feedback about themselves across contexts.

These aims align with established metacognitively oriented interventions, though to differing degrees. Metacognitive reflection and insight therapy (MERIT) explicitly targets self-reflectivity and the capacity to form increasingly integrated representations of oneself and others, mechanisms that correspond closely to the self-referential processing that had the strongest association with insight in the present review (Lysaker et al., 2018). By contrast, metacognitive training (MCT) primarily targets domain-specific cognitive biases, such as jumping to conclusions and overconfidence (Moritz and Woodward, 2007); these processes showed little consistent association with insight here. This distinction suggests that interventions emphasizing integrative self-reflection may be particularly well suited to targeting the processes most closely tied to insight, whereas approaches focused primarily on correcting discrete reasoning biases may act on processes less directly related to it (see Lopez-Morinigo et al., 2020, for review). Overall, interventions whose implementers aim to improve insight may benefit from targeting self-referential processing beyond domain-specific cognitive biases.

We also recognize several limitations of the pooled synthesis. The number of studies per subgroup was small ($k = 3$ per pooled subgroup), limiting precision and statistical power, particularly for Meta-WCST variants where between-study heterogeneity was substantial. Three task-type subgroups (probabilistic reasoning, metamemory, social metacognition) contained only one study each and could not be pooled. Many primary studies reported multiple correlations across insight subscales and metacognitive indices; representative effect sizes were selected on a pre-specified basis aligned with each study's primary task

category, but a sensitivity analysis using alternative effect-size choices was not performed because of the small number of studies. Egger's test was significant; however, the asymmetry is largely explained by the systematic relationship between task type and effect size, as small-N studies in the review tended to employ self-reflection or Meta-WCST paradigms (yielding moderate effects), while larger-N studies tended to employ perceptual, probabilistic, or social paradigms (yielding near-zero effects). The small k further limits the reliability of bias-assessment procedures. Notably, the strongest finding of the synthesis, that task type accounts for the majority of between-study heterogeneity, is not affected by these caveats, and the pooled pattern converges with the narrative synthesis in supporting the hierarchical interpretation. An additional consideration is that clinical and cognitive insight measures were combined within the pooled analyses. As noted in the Methods, the two share a common self-reflective core and are frequently assessed together, which motivated their combination. Clinical and cognitive insight do nonetheless differ conceptually, and the narrative synthesis suggests that they may be differentially associated with task-based metacognition. The pooled estimate is therefore best read as an aggregate across insight measures rather than as specific to either construct. The available number of studies did not enable separate meta-analyses by insight type without compromising the task-type subgrouping that constituted the primary analysis and disaggregated meta-analysis by insight type will be an informative direction for future work as the evidence base expands.

Additionally, we did not use a formal graded risk-of-bias instrument; although all studies used validated measures, varied chiefly in precision, and had their key design features appraised and tabulated, the absence of a standardized quality rating should be borne in mind when interpreting the pooled estimates. Finally, although clinical high-risk samples were eligible, none of the included studies evaluated individuals at clinical high risk for psychosis, so our conclusions are restricted to established schizophrenia-spectrum disorders. Together with the small number of studies per diagnostic category, this precluded an examination of clinical group as a moderator (e.g., at-risk versus established illness or between established diagnoses). Whether task-based metacognition relates to insight earlier in the psychosis continuum (i.e., before illness chronicity and treatment exposure) remains an open and important question and a priority for future research.

This systematic review examined the relationship between task-based metacognition and insight in schizophrenia spectrum disorders, finding that the association depends critically on how researchers operationalize metacognition. Perceptual decision-making, probabilistic reasoning, and metamemory tasks showed no consistent relationship with insight, whereas Meta-WCST and self-reflection paradigms demonstrated more meaningful associations, particularly with clinical insight. These findings suggest that task-based metacognition and insight are partially dissociable constructs, with impaired insight reflecting a specific difficulty with integrative self-reflection rather than a general metacognitive deficit, implying that recognizing one's illness depends less on evaluating discrete cognitive performances and more on forming coherent, contextualized representations of oneself over time. In future researchers should employ longitudinal designs, compare paradigms within the same sample, and extend findings to early psychosis populations, while intervention approaches may benefit from prioritizing self-reflective capacities over domain-general metacognitive accuracy.

CRedit authorship contribution statement

Asala Halaj: Writing – original draft, Methodology, Formal analysis, Data curation. **Tricia X.F. Seow:** Writing – review & editing. **Anthony S. David:** Writing – review & editing, Formal analysis, Conceptualization.

Role of the funding source

No funding source had any role in the design of the study, data collection, analysis, or writing of the manuscript.

Declaration of competing interest

The authors declare no conflicts of interest.

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